REAL-TIME ADAPTIVE NOISE CANCELLATION USING DUAL MICROPHONES IN MATLAB

ABSTRACT

In real-world audio communication systems, background noise significantly degrades speech quality. Adaptive Noise Cancellation (ANC) is an effective signal processing technique that suppresses unwanted noise using a reference noise signal. This project presents the design and implementation of a **real-time adaptive noise cancellation system** using **two microphones** and **MATLAB**. A **Normalized Least Mean Square (NLMS)** adaptive filter is employed to estimate and subtract correlated noise from the primary speech signal. Real-time audio acquisition, adaptive filtering, and live audio playback are successfully demonstrated. Experimental results show effective reduction of background noise while preserving speech intelligibility, validating the effectiveness of the proposed system.

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1. INTRODUCTION

Noise is an unavoidable component in practical audio environments such as classrooms, offices, and industrial spaces. Conventional fixed filters fail to adapt to time-varying noise characteristics. Adaptive Noise Cancellation (ANC) overcomes this limitation by continuously updating filter coefficients based on the input signals.

ANC systems are widely used in applications such as:

- Headphones
- Communication systems
- Automotive cabins
- Industrial monitoring systems

This project focuses on implementing a **real-time ANC system** using MATLAB, emphasizing **correct DSP theory**, **live signal processing**, and **practical constraints**.

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2. PROBLEM STATEMENT

To design and implement a **real-time adaptive noise cancellation system** that reduces background noise from live audio using:

- A **primary microphone** capturing speech with noise
- A **reference microphone** capturing correlated noise

The system must operate in real time and demonstrate measurable noise reduction.

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3. SYSTEM OVERVIEW

3.1 System Architecture

The system follows a **feedforward ANC structure**:

- **Primary Microphone (Laptop Mic)**
  Captures desired speech signal contaminated with noise.
- **Reference Microphone (Mobile Phone Mic via WO Mic)**
  Captures background noise correlated with the noise present in the primary signal.
- **Adaptive Filter (NLMS)**
  Learns the noise path and estimates noise contribution.
- **Subtractor**
  Removes estimated noise from the primary signal.
- **Output**
  Noise-reduced speech played through headphones.

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4. ALGORITHM USED

4.1 Normalized Least Mean Square (NLMS) Algorithm

The NLMS algorithm is an improvement over the standard LMS algorithm. It normalizes the step size with respect to the input signal power, providing:

- Faster convergence
- Improved numerical stability
- Robust performance under varying signal amplitudes

The adaptive filter minimizes the error signal:

$$e(n) = x(n) - \hat{y}(n)$$

where:

- $x(n)$= primary signal (speech + noise)
- $\hat{y}(n)$= estimated noise
- $e(n)$= noise-reduced output

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5. IMPLEMENTATION DETAILS

5.1 Software Tools

- MATLAB R2023a
- Audio Toolbox
- DSP System Toolbox

5.2 Hardware Setup

- Laptop with built-in microphone (Primary mic)
- Smartphone microphone via WO Mic (Reference mic)
- USB data cable
- Headphones (to avoid acoustic feedback)

5.3 Key Parameters

- Frame length: 256 samples
- Adaptive filter length: 64 taps
- NLMS step size: 0.12–0.15 (optimized experimentally)

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6. REAL-TIME OPERATION

The system operates in two modes:

1. **Live ANC Mode**
   Real-time noise cancellation with audible output through headphones.
2. **Analysis Mode**
   Finite-duration signal capture used for generating plots and performance evaluation.

A buffer-priming stage is used to avoid initial silent frames from audio drivers.

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7. RESULTS AND DISCUSSION

7.1 Time-Domain Analysis

Time-domain plots show a visible reduction in noise amplitude after adaptive filtering while preserving speech components.

7.2 Frequency-Domain Analysis

FFT magnitude spectra indicate attenuation of dominant noise frequency components in the ANC output signal.

7.3 Energy Reduction

Frame-wise energy plots confirm that the output signal energy is consistently lower than the input signal energy, demonstrating effective noise cancellation.

7.4 Subjective Listening Test

Listening tests using headphones confirm:

- Reduced background noise
- Clear speech reproduction
- Stable performance without oscillations

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8. LIMITATIONS

- ANC cancels only **correlated noise** present in both microphones.
- Sudden or uncorrelated noise cannot be fully eliminated.
- Different microphone clock sources may introduce minor drift.
- MATLAB implementation is floating-point and not optimized for embedded deployment.

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9. FUTURE WORK

- Embedded implementation using DSP or microcontroller platforms

- Fixed-point optimization for real-time hardware deployment
- Custom PCB with MEMS microphones
- Integration with Bluetooth or wireless audio systems

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10. CONCLUSION

This project successfully demonstrates a **real-time adaptive noise cancellation system** using MATLAB and dual microphones. The NLMS adaptive filter effectively reduces background noise while maintaining speech quality. Both quantitative analysis and subjective listening tests validate the system's performance. The project provides a strong foundation for further development toward embedded and industrial ANC applications.

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